CADSNET

Use case diagram and gameplay flow by attack

Based on the game build of 3 November 2026.

1. Use case diagrams

The Player (Alice, the student) is the main actor. The Attacker (the Ghostline group) is the second actor and only takes part in the parts of the game where a cyber threat happens. Solid arrows connect an actor to a use case; dashed arrows marked «include» or «extend» connect a use case to the steps it includes or the options it can lead to.

System Use Case Diagram

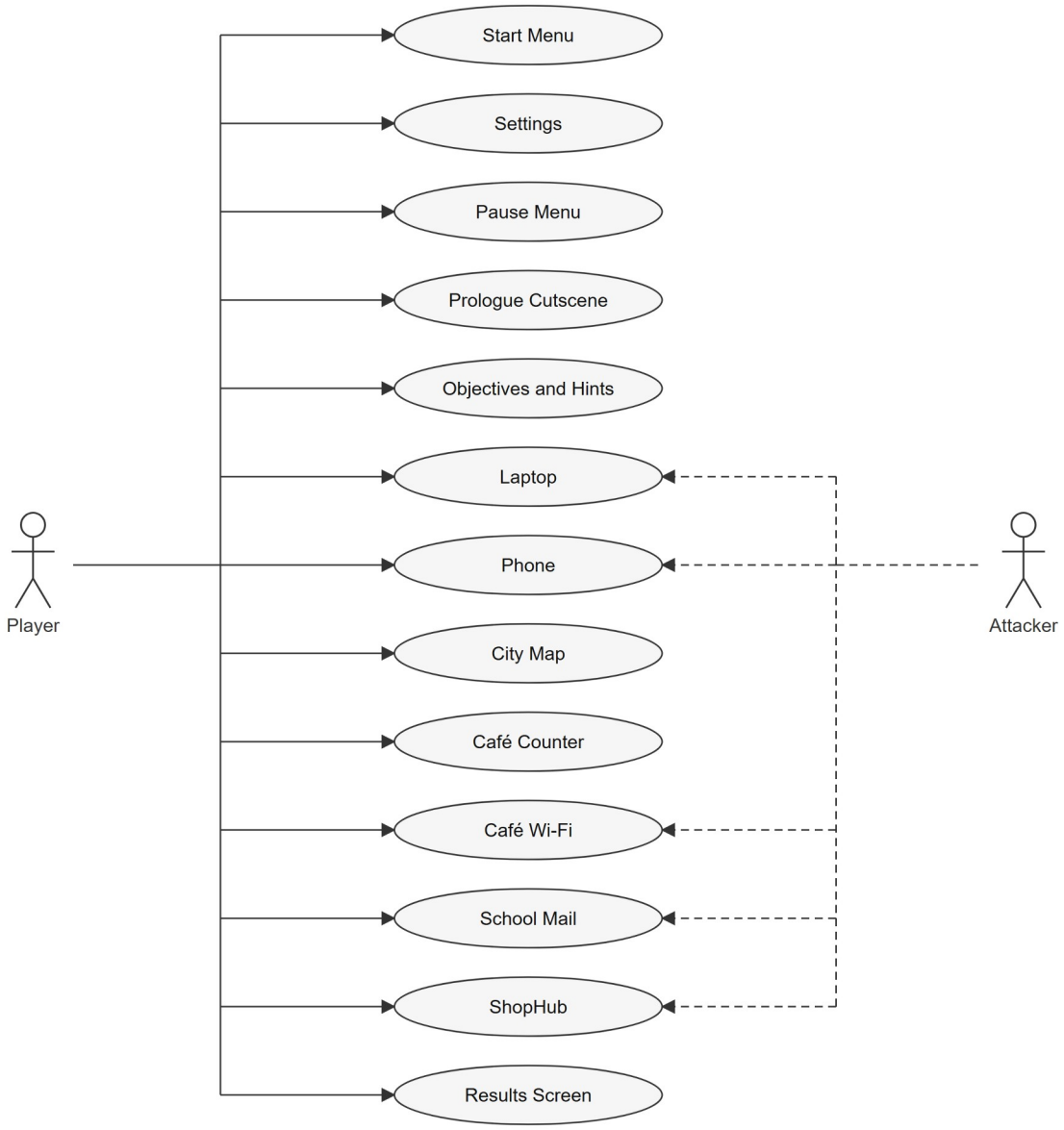


Figure 4.1 System Use Case Diagram

Use Case Diagram of Start Menu

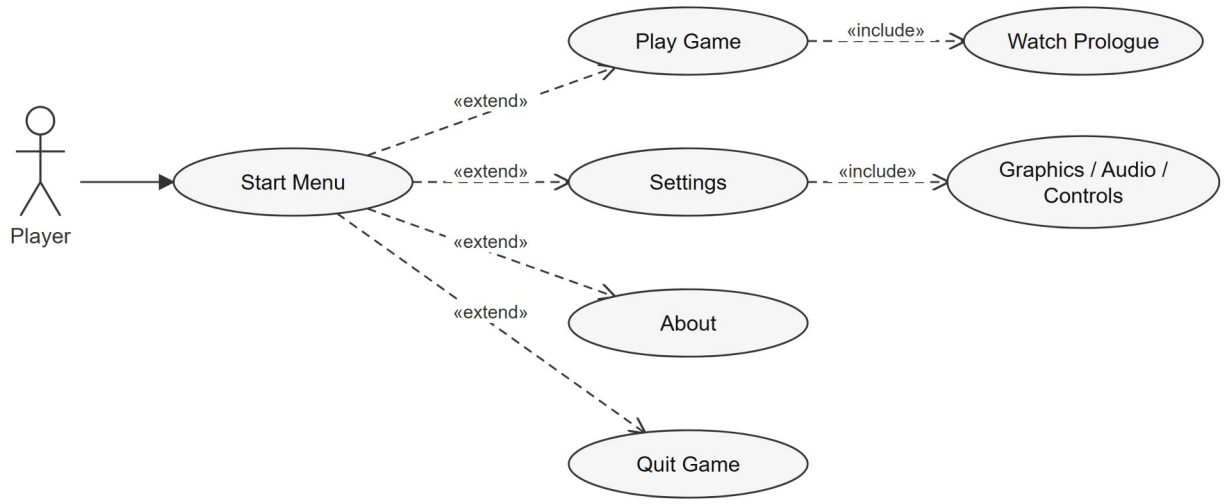


Figure 4.2 CN001 Use Case Diagram of Start Menu

Use Case Diagram of Back Up the Thesis (Laptop and Flash Drive)

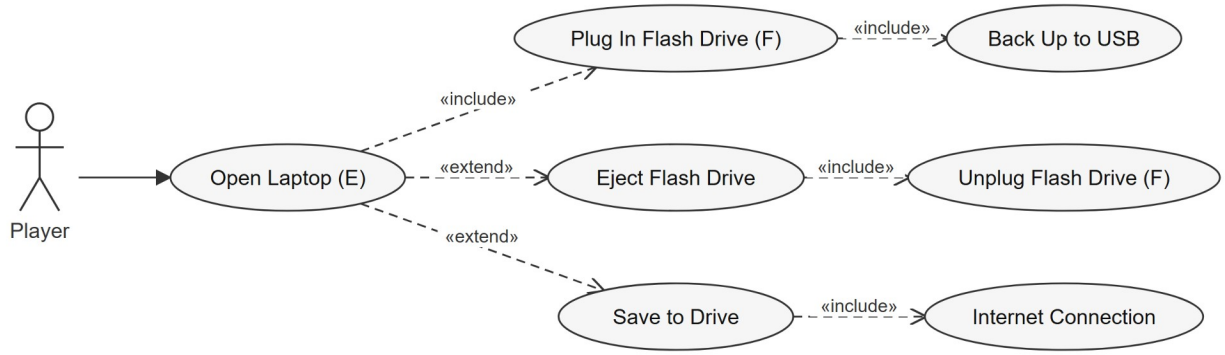


Figure 4.3 CN002 Use Case Diagram of Back Up the Thesis (Laptop and Flash Drive)

Use Case Diagram of Turn On 2FA (Phone)

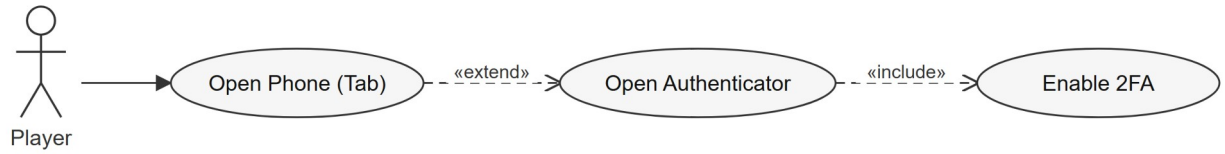


Figure 4.4 CN003 Use Case Diagram of Turn On 2FA (Phone)

Use Case Diagram of Connect to the Café Wi-Fi

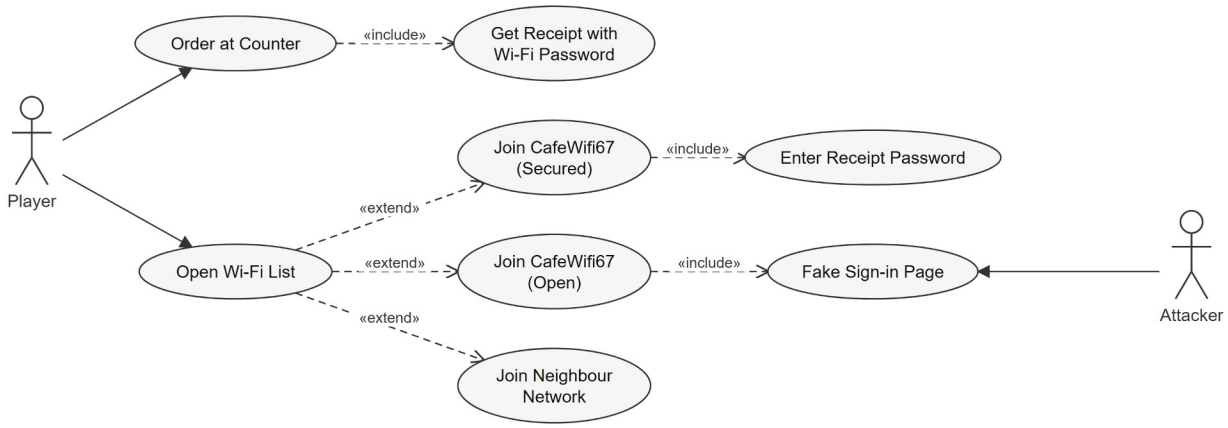


Figure 4.5 CN004 Use Case Diagram of Connect to the Café Wi-Fi

Use Case Diagram of Send the Thesis (School Mail)

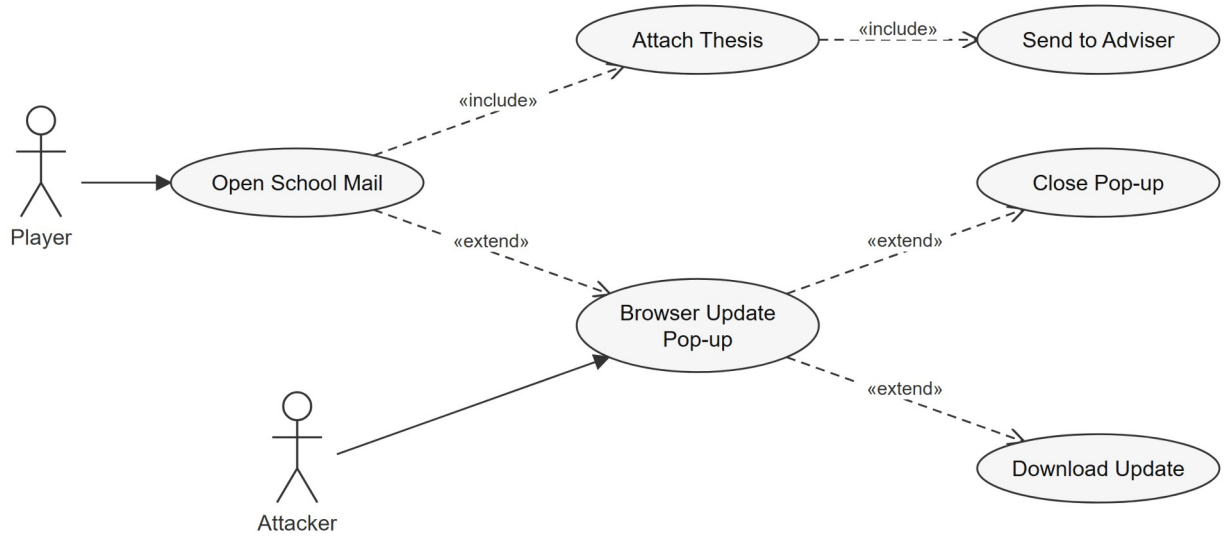


Figure 4.6 CN005 Use Case Diagram of Send the Thesis (School Mail)

Use Case Diagram of Sign-in Alert

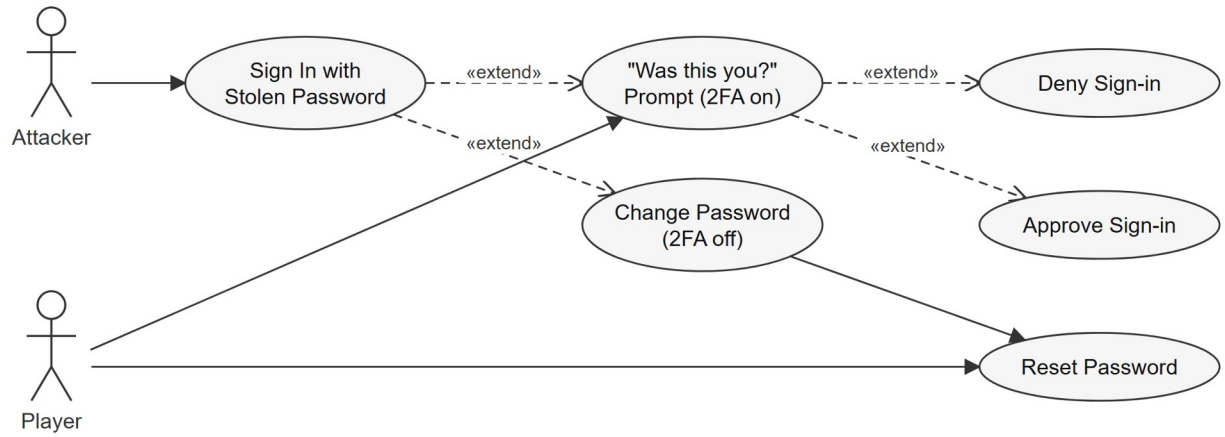


Figure 4.7 CN006 Use Case Diagram of Sign-in Alert

Use Case Diagram of Kai's Money Request

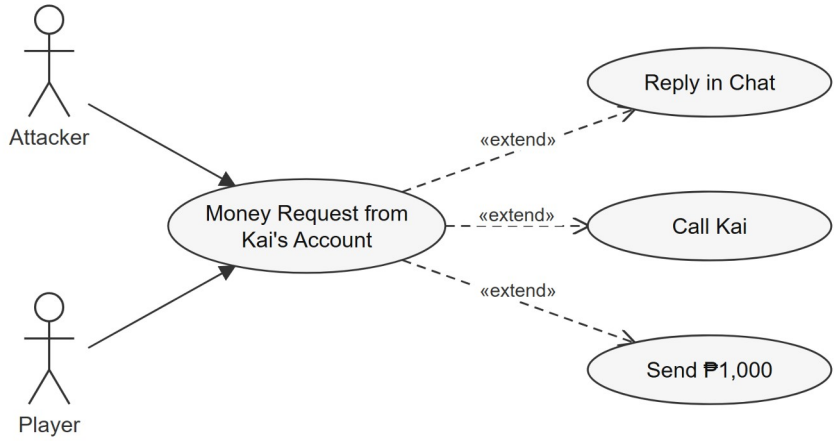


Figure 4.8 CN007 Use Case Diagram of Kai's Money Request

Use Case Diagram of Locked Files (Ransomware)

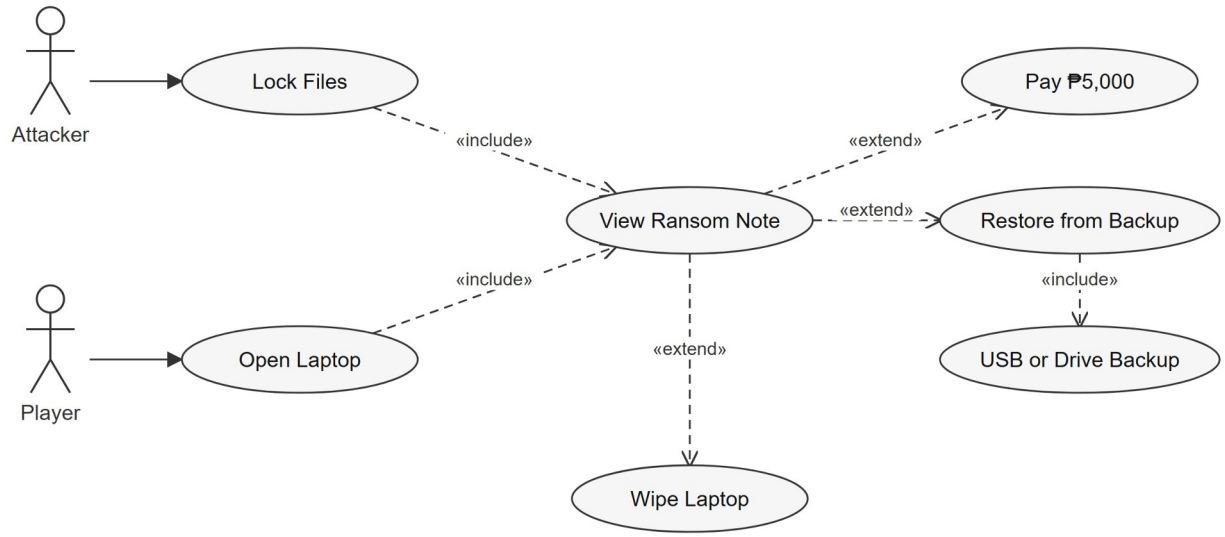


Figure 4.9 CN008 Use Case Diagram of Locked Files (Ransomware)

Use Case Diagram of ShopHub

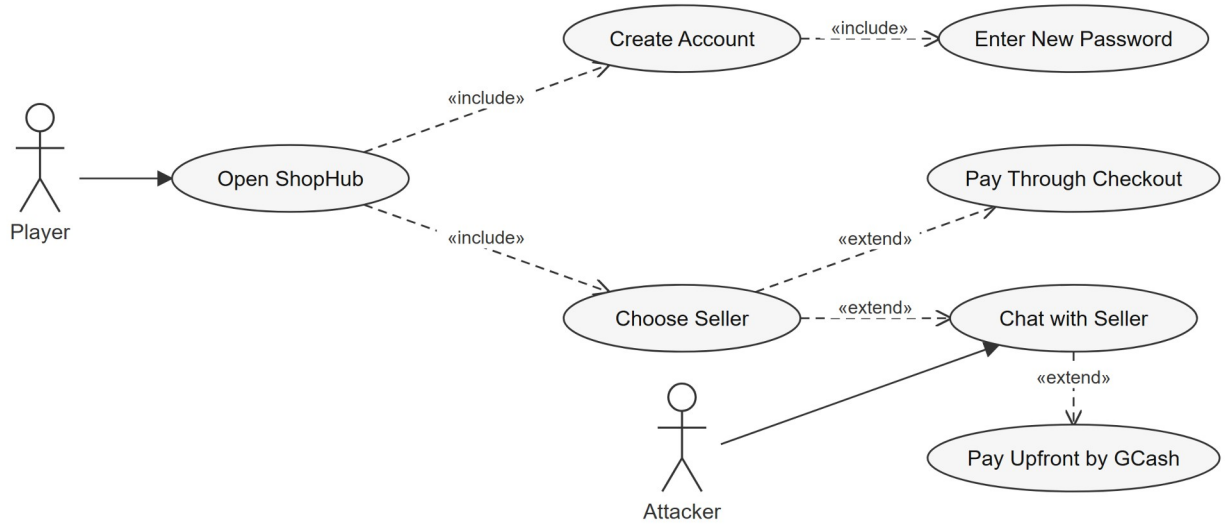


Figure 4.10 CN009 Use Case Diagram of ShopHub

Use Case Diagram of Results and Ending

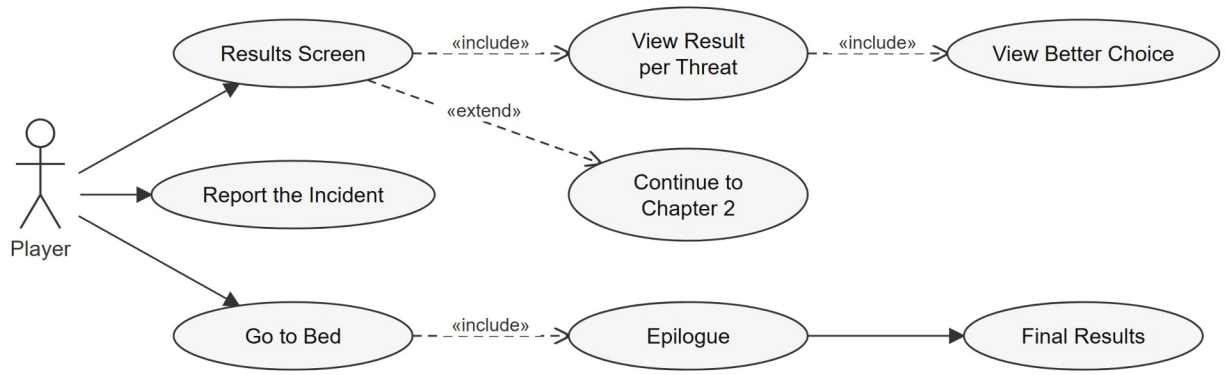


Figure 4.11 CN010 Use Case Diagram of Results and Ending

2. Story order

The game is played in this order. Each attack in section 3 happens inside one of these segments.

Segment	Where	Attacks and habits
Prologue	Alice's room, 11:47 PM	Story setup: Kai's texts, Ghostline on the TV news
Chapter 1 · Home	House, 9:04 AM	Back up your files, Two-factor authentication
Chapter 1 · Café	City and café	Fake café Wi-Fi (evil twin), Fake browser update; thesis is sent
Chapter 1 results	Results screen	Safe / Recovered / Compromised per item, with the better choice
Chapter 2 · That night	House, 11:40 PM	Stolen password used to sign in, Friend's hacked account, Files locked for ransom, Reused password, Fake online seller
Ending	Laptop and bed	Report the incident, "Two weeks later..." epilogue, final results

3. Gameplay flow by attack

Each part below covers one attack or habit: a table with where it happens, what the player sees and each result, then a flowchart of the choices on the next page. The colours in the tables and charts mean:

- **Green, Safe:** the right choice.
- **Amber, Recovered:** a mistake that was caught or fixed in time.
- **Red, Compromised:** the attack worked.
- **Grey dashed box:** an effect that shows up later in the story.

Some attacks depend on earlier ones: the stolen login comes from the fake Wi-Fi, the ransomware from the fake update, and 2FA and the backup decide how bad those get.

3.1 Back up your files

Where	Chapter 1 · Home (USB) and café (Drive)
Attack	Data loss (prepares for ransomware)
What the player sees	The thesis can't be sent at home because there is no internet. The flash drive is in the living room; F plugs it into the laptop. Back-up to USB copies the thesis; Eject in the taskbar, then F, unplugs it. Save to Drive only works once online at the café.
Safe	Two backups: the flash drive and the school Drive.
Recovered	One backup. Better than none, but one lost or locked copy means the work is gone.
Compromised	Left home with a single copy of the thesis.
Later effect	In Chapter 2, a backup lets the player restore the thesis after ransomware. Without one, wiping the laptop loses it.
Lesson	Keep at least two backups in different places: one offline (a flash drive you unplug) and one in the cloud.
Source	CISA, #StopRansomware Guide (2023): maintain offline, encrypted backups

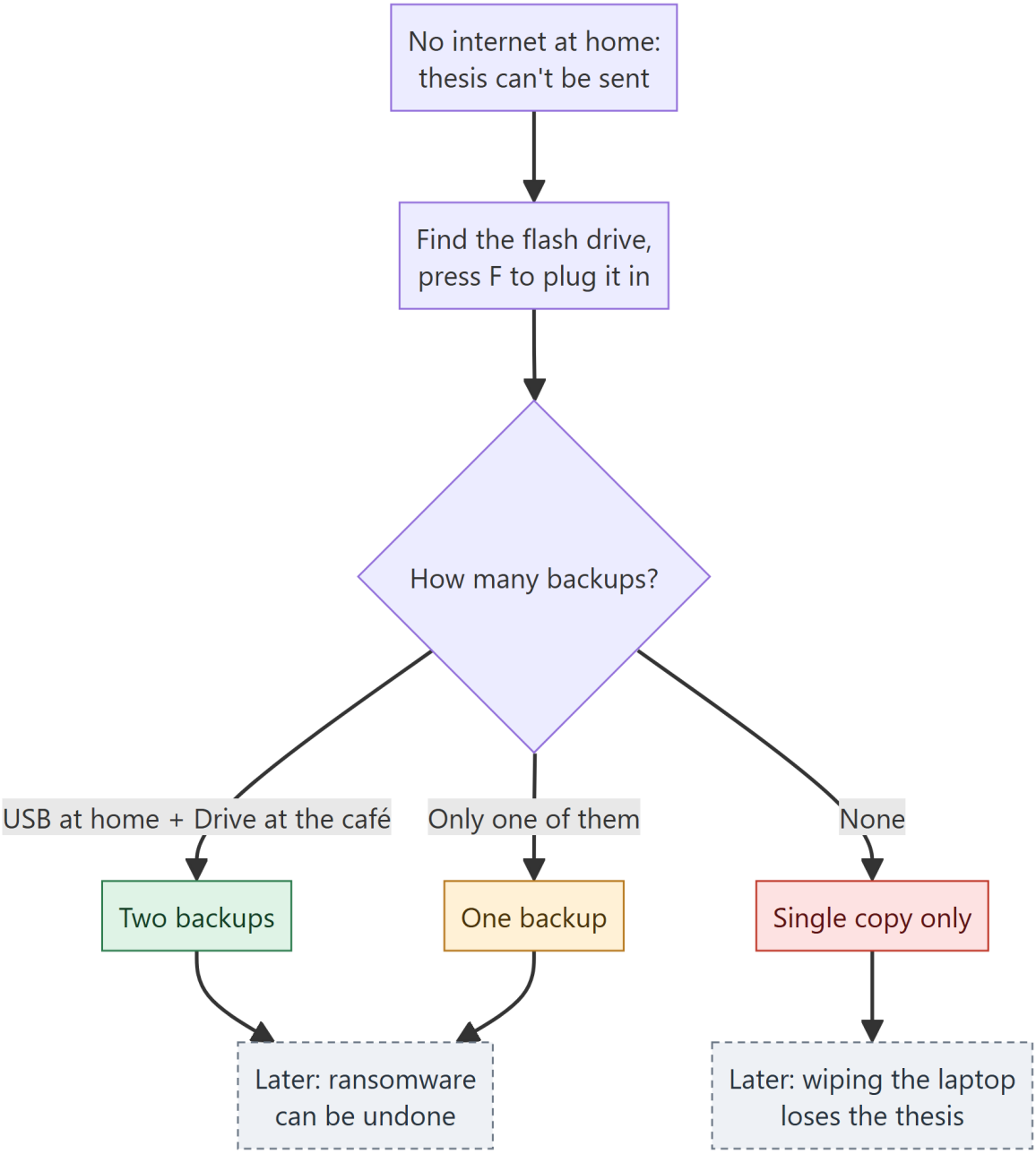


Figure 4.12 Flowchart of Back up your files

3.2 Two-factor authentication

Where	Chapter 1 · Home, after the backup
Attack	Account takeover (prepares for the stolen login)
What the player sees	Alice says she could turn on 2FA on her phone. The phone offers to set up the authenticator.
Safe	2FA on: a stolen password alone is not enough to get into the account.
Compromised	The account is protected by a password only.
Later effect	Decides what happens in the Chapter 2 sign-in alert and the ShopHub password leak.
Lesson	Turn on MFA for school, email and banking accounts.
Source	CISA, Secure Our World: "Turn On MFA"; Microsoft Security Blog (Aug 2019)

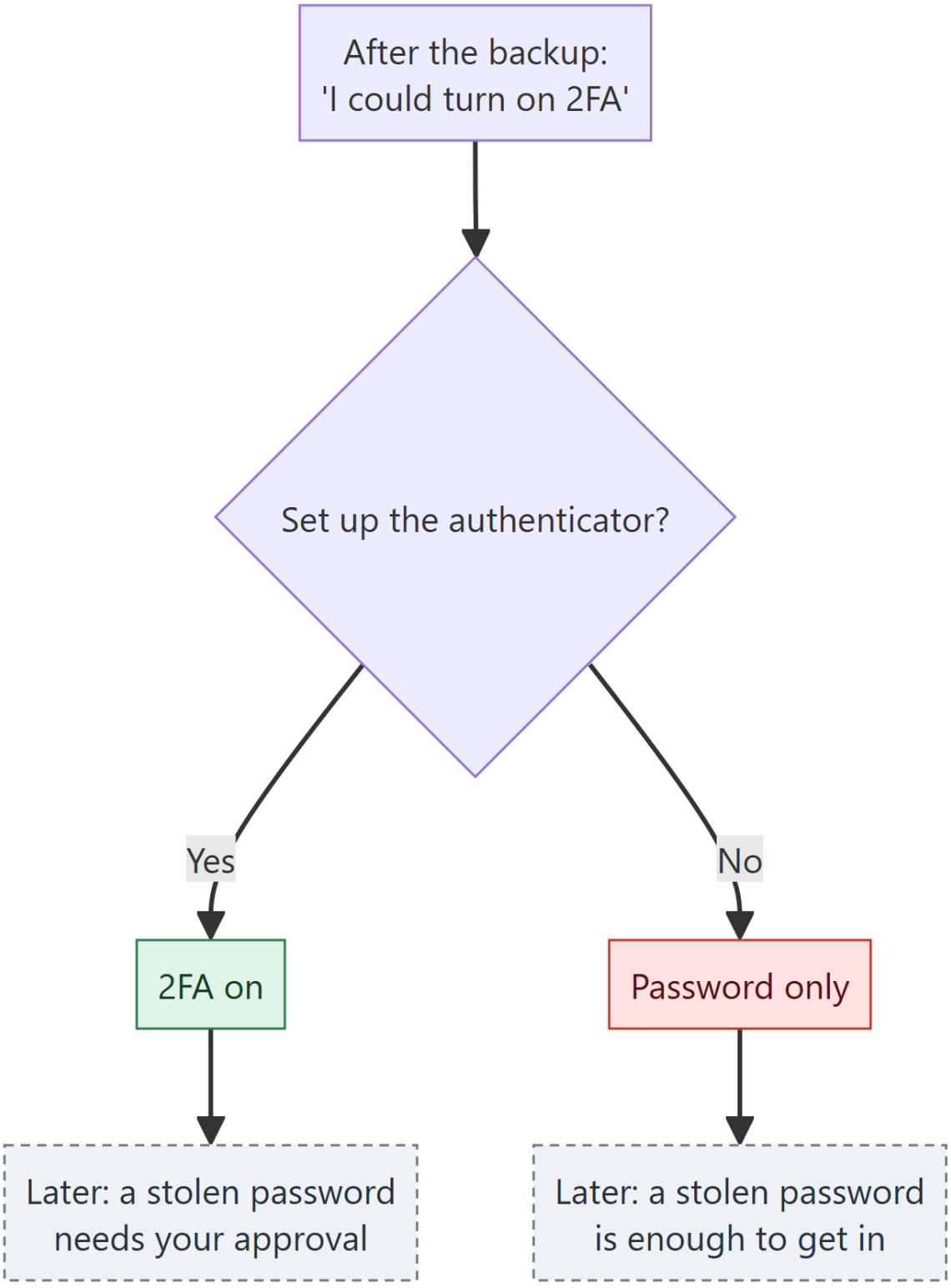


Figure 4.13 Flowchart of Two-factor authentication

3.3 Fake café Wi-Fi (evil twin)

Where	Chapter 1 · Café
Attack	Evil twin access point with a fake sign-in page (credential theft)
What the player sees	Three networks: CafeWifi67 (Secured), CafeWifi67 (Open) and the neighbour's PLDT_HOME_7A2C. The café receipt shows the real password. The open copy opens a sign-in page asking for the school email and password.
Safe	Joined the secured CafeWifi67 with the receipt password.
Recovered	Opened the open copy's sign-in page but backed out without typing the login.
Compromised	Typed the school email and password into the fake sign-in page.
Later effect	A stolen login triggers the sign-in alert in Chapter 2.
Lesson	Attackers copy a café's exact Wi-Fi name. Join the network that asks for the password on your receipt, and never type your email password into a Wi-Fi sign-in page.
Source	FTC, "Are Public Wi-Fi Networks Safe? What You Need To Know" (2023)

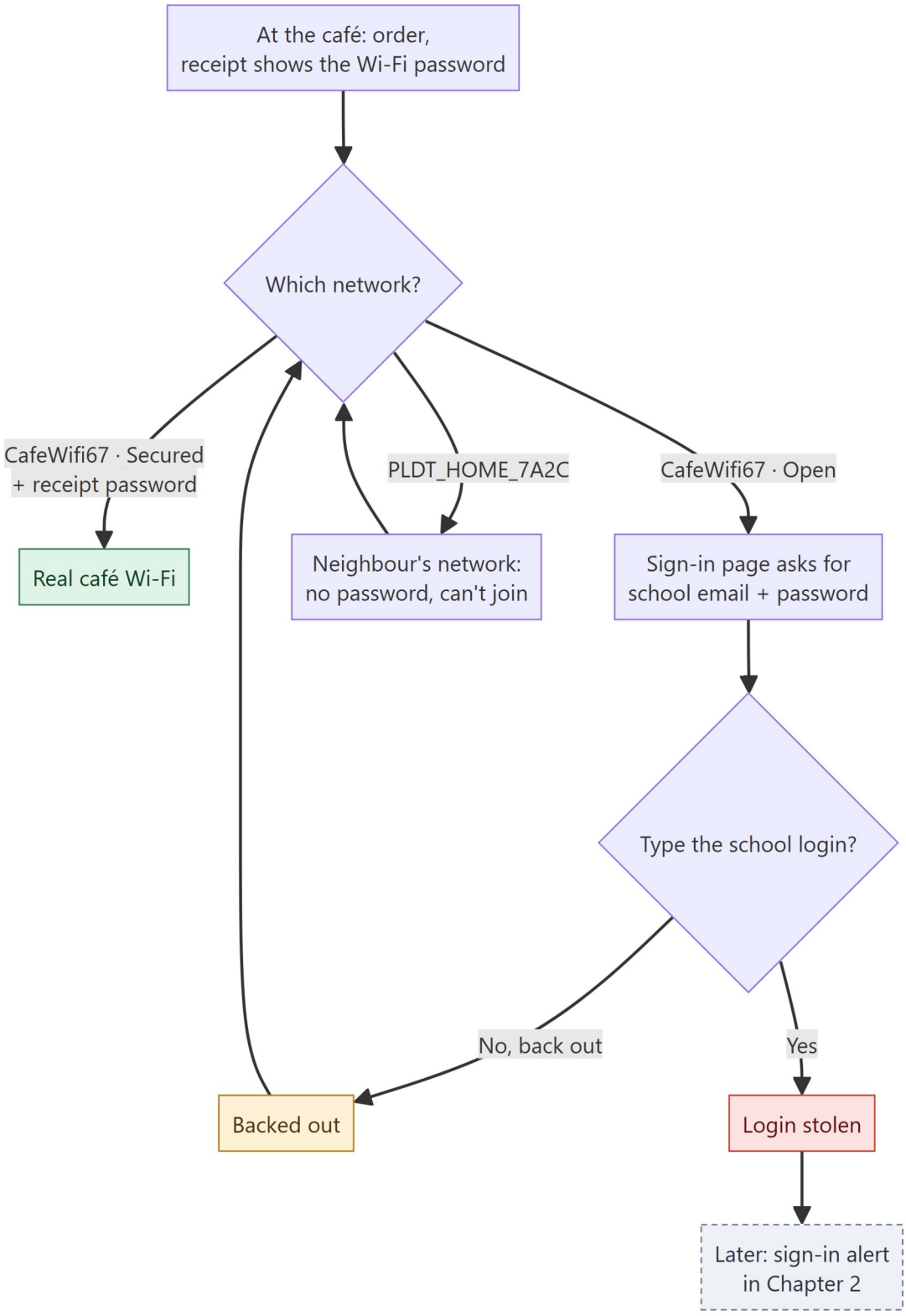


Figure 4.14 Flowchart of Fake café Wi-Fi (evil twin)

3.4 Fake browser update

Where	Chapter 1 · Café, when the school mail page loads
Attack	Malware delivered as a fake update (leads to ransomware)
What the player sees	A pop-up in the browser says the browser is out of date, with X and Download.
Safe	Closed the pop-up; nothing was installed.
Compromised	Downloaded the "update" from the web page.
Later effect	The download was ransomware: in Chapter 2 the laptop's files are locked.
Lesson	Only update software through your system's own updater or the developer's official site.
Source	CISA, Secure Our World: "Update Software"

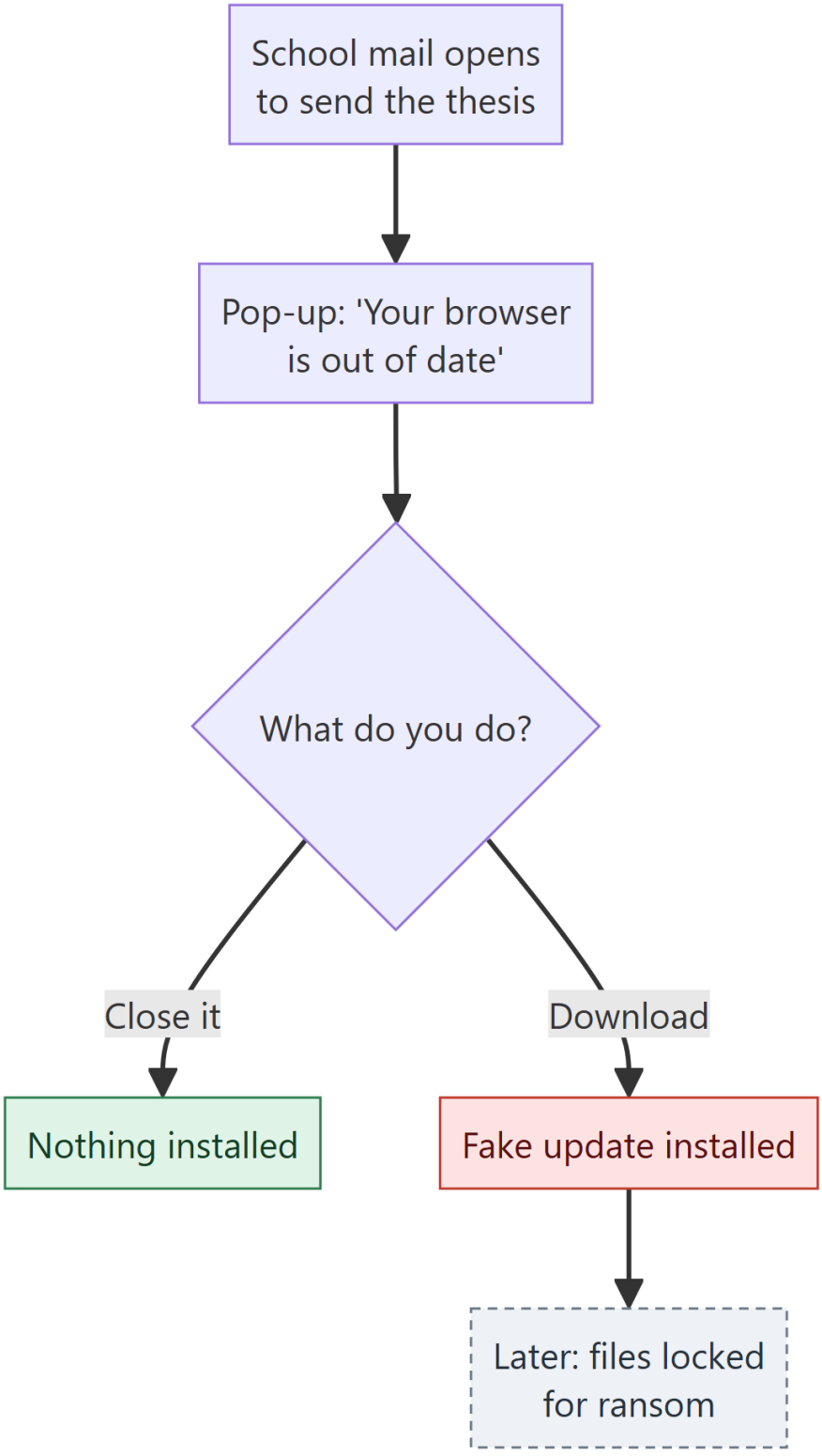


Figure 4.15 Flowchart of Fake browser update

3.5 Stolen password used to sign in

Where	Chapter 2 · Phone, only if the login was stolen at the café
Attack	Account takeover with a stolen password
What the player sees	With 2FA: a "Was this you?" prompt with a 10-second timer. Without 2FA: a "Your password was changed" notice.
Recovered	2FA on and the player pressed Deny: the attacker stayed out. Change the password.
Compromised	Approved the sign-in, let the timer run out, or had no 2FA.
Lesson	Never approve a sign-in request you didn't start: deny it, then change your password.
Source	CISA, "Implementing Number Matching in MFA Applications" (2022)

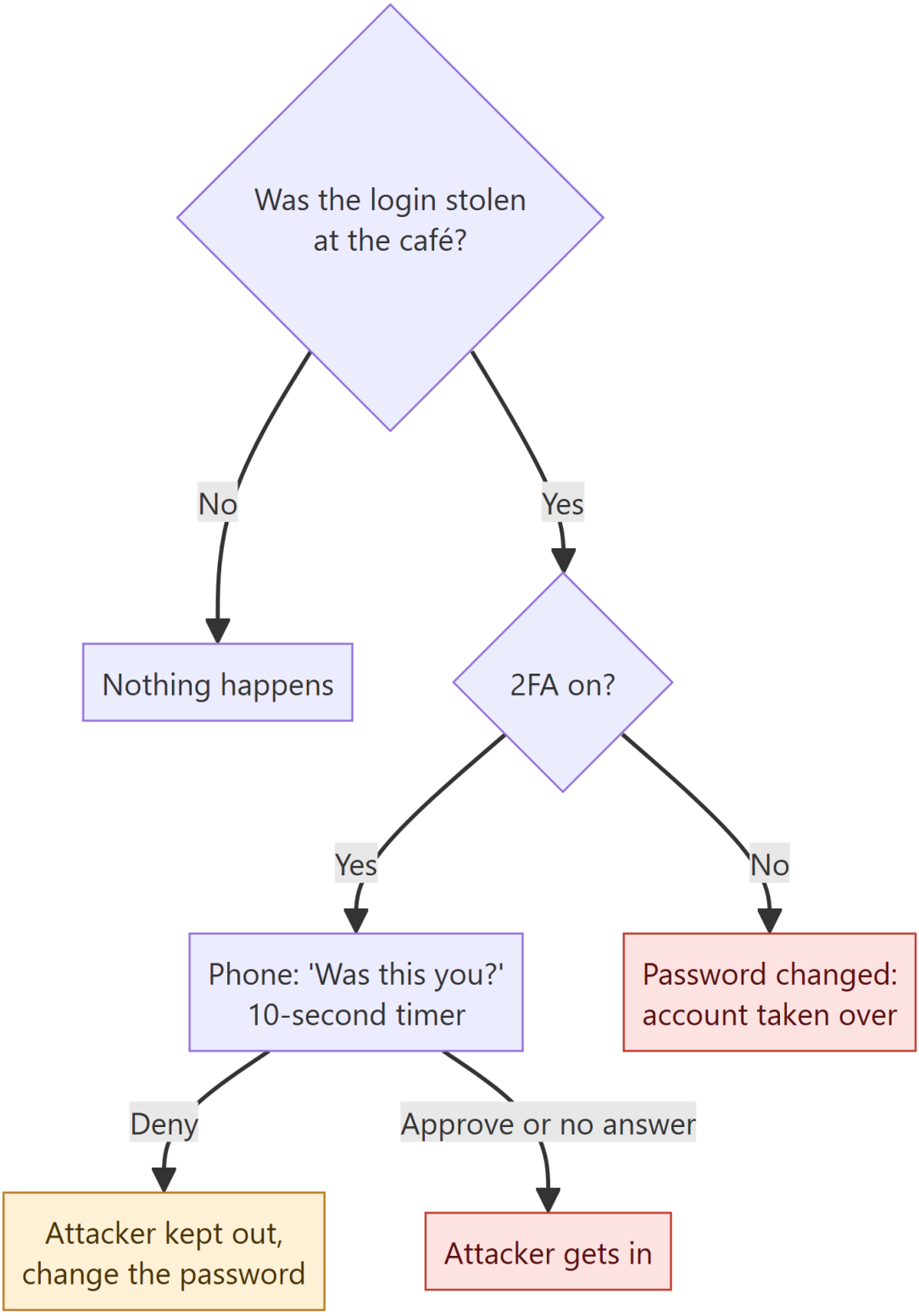


Figure 4.16 Flowchart of Stolen password used to sign in

3.6 Friend's hacked account asks for money

Where	Chapter 2 · Phone messages
Attack	Impersonation through a hijacked friend's account
What the player sees	Kai's account asks for ₱1,000. The player can reply in the chat, call Kai, or send the money.
Safe	Called Kai, learned the account was hacked, sent nothing.
Compromised	Sent ₱1,000 to the scammer.
Later effect	Replying in the same chat only reaches the scammer ("yes it's me"), so it proves nothing and the choice stays open.
Lesson	When a friend's account suddenly asks for money, verify through another channel, like a phone call.
Source	FTC, "How to Recover Your Hacked Email or Social Media Account"

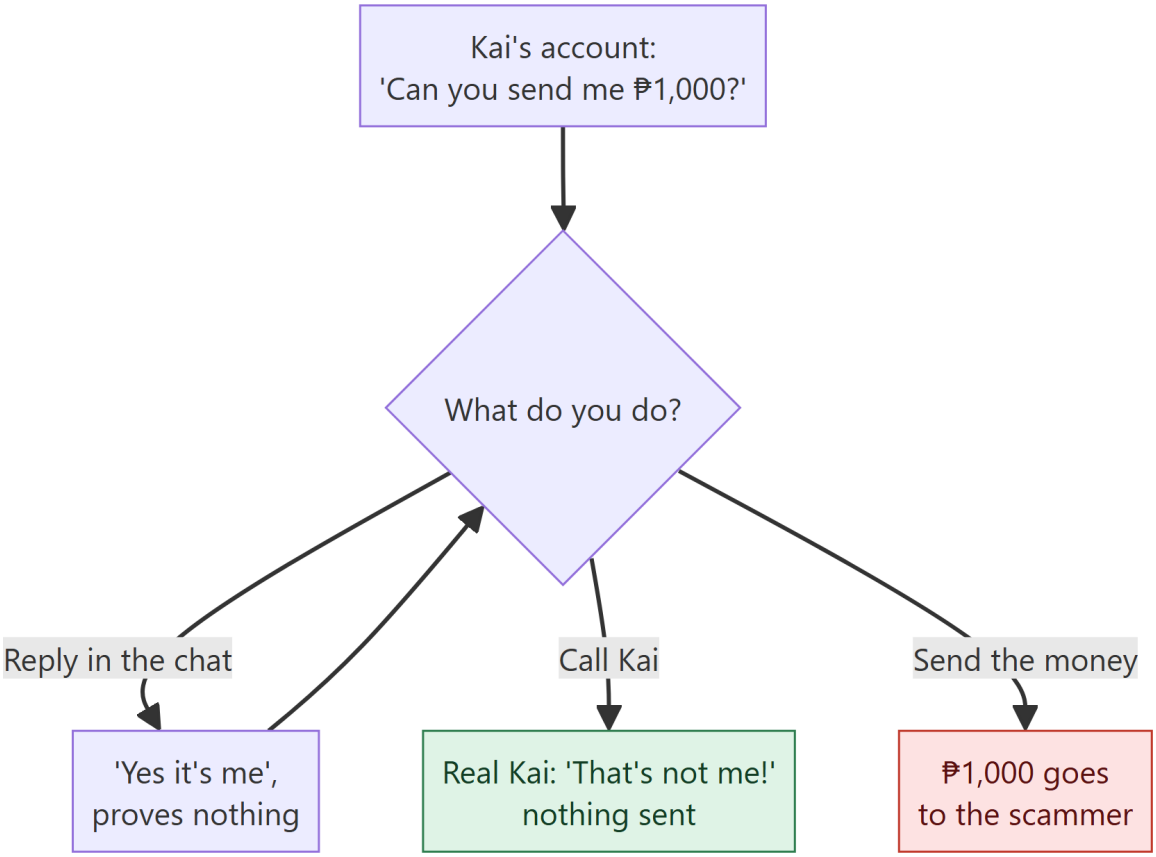


Figure 4.17 Flowchart of Friend's hacked account asks for money

3.7 Files locked for ransom

Where	Chapter 2 · Laptop, only if the fake update was installed
Attack	Ransomware
What the player sees	The laptop's files are locked with a demand for ₱5,000. Choices: pay, restore from backup (only with a backup), or wipe the laptop.
Safe	Didn't pay: restored the thesis from the USB or Drive backup.
Recovered	Didn't pay: wiped the laptop with no backup, so the thesis is lost.
Compromised	Paid. The attackers asked for more and nothing unlocked.
Lesson	Never pay a ransom. An offline backup is what lets you recover.
Source	CISA, #StopRansomware Guide (2023)

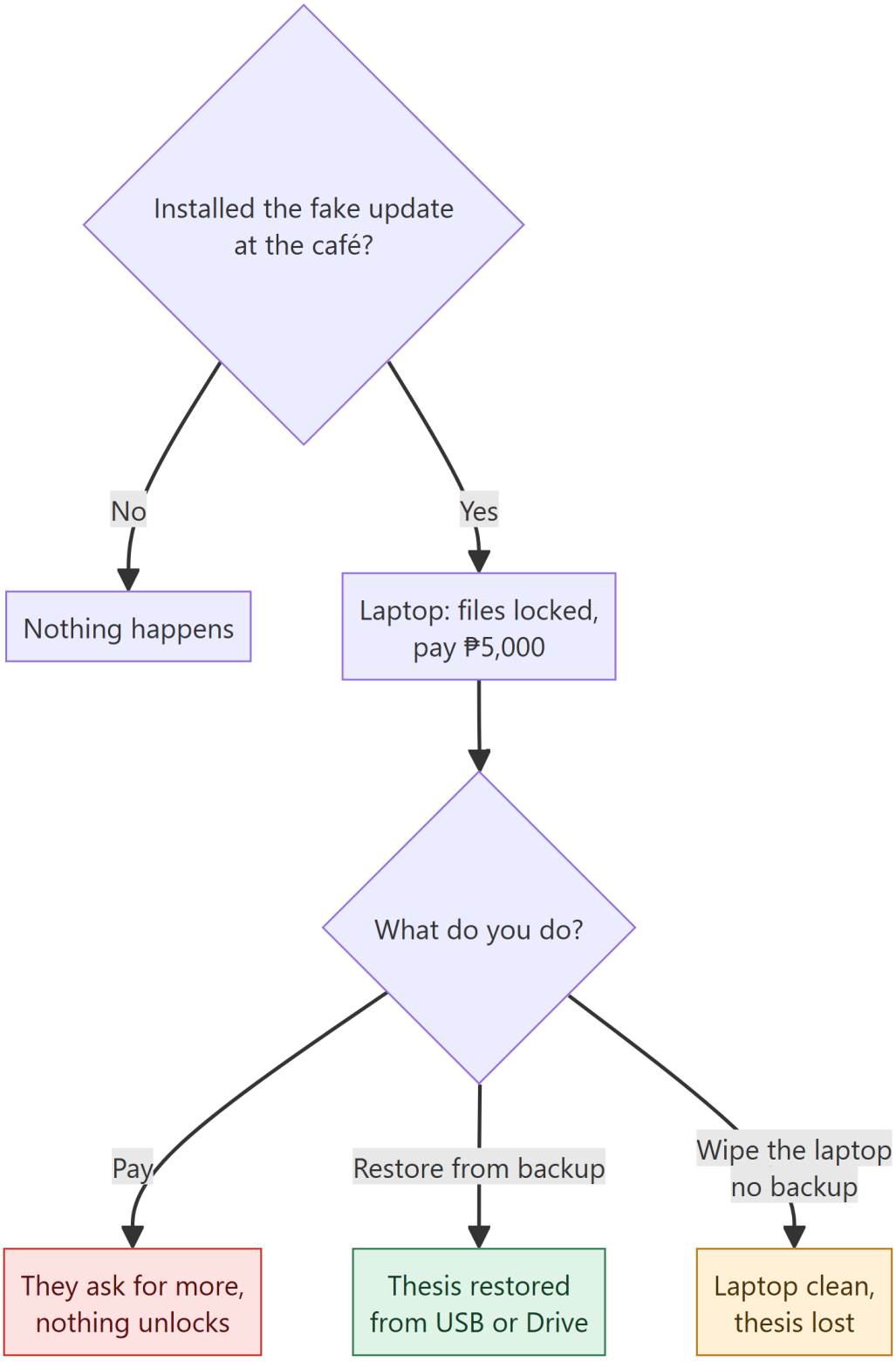


Figure 4.18 Flowchart of Files locked for ransom

3.8 Reused password

Where	Chapter 2 · Laptop, ShopHub sign-up
Attack	Password reuse exposed by a third-party breach
What the player sees	ShopHub asks for a new account password. The player can make a new long one or reuse the school password.
Safe	Made a new, long password just for ShopHub.
Compromised	Reused the school password.
Later effect	After the laptop closes, ShopHub is breached and the school account is attacked again: with 2FA the player gets a "Was this you?" prompt to deny, without it the account is taken over.
Lesson	Use a different, long password for every account (CISA recommends at least 16 characters). A password manager remembers them.
Source	CISA, Secure Our World: "Use Strong Passwords"

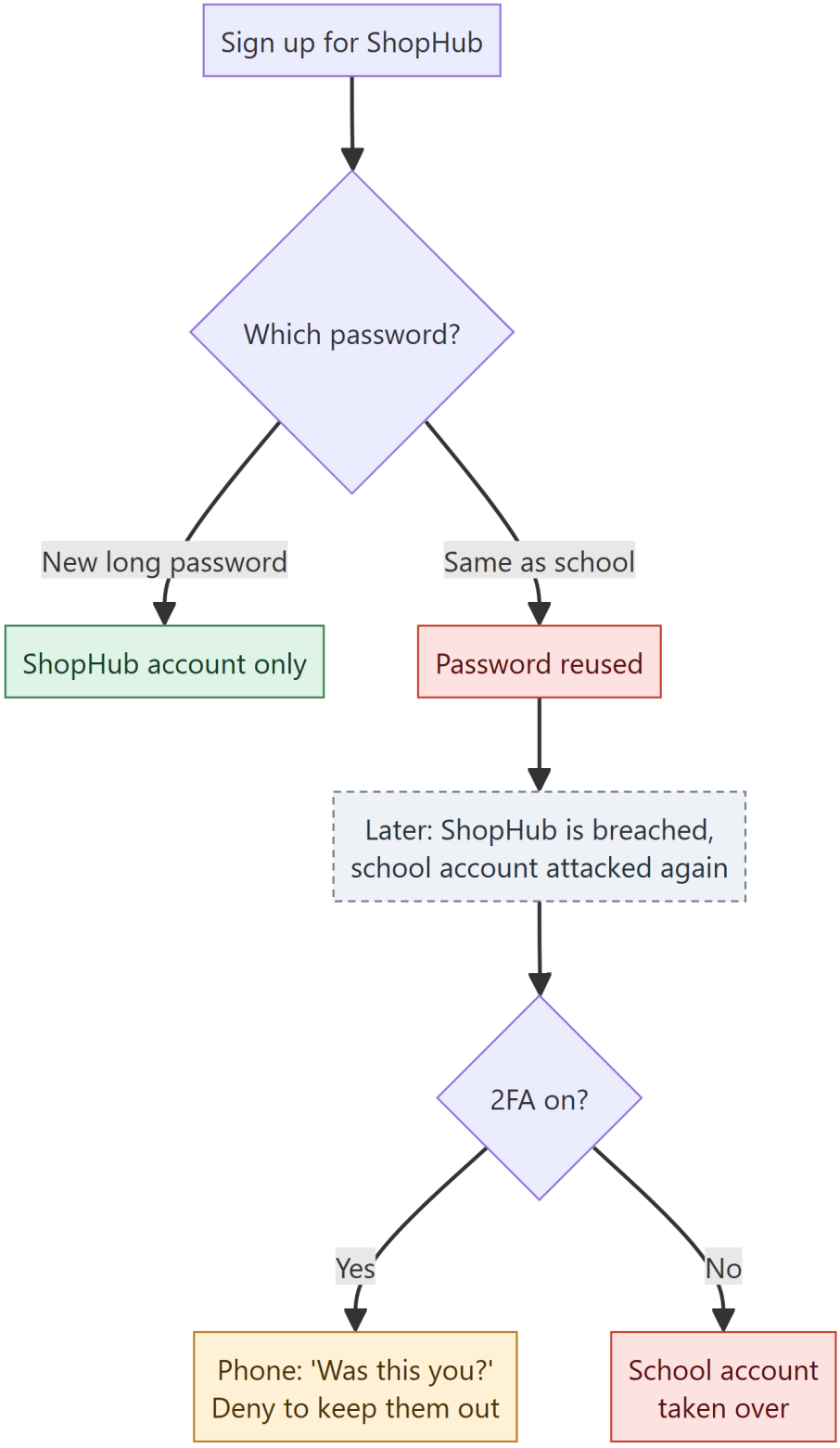


Figure 4.19 Flowchart of Reused password

3.9 Fake online seller

Where	Chapter 2 · Laptop, ShopHub
Attack	Online selling scam
What the player sees	Buying a calculator. The official store has years of good reviews; gadget.deals.ph is new, has no reviews, is far too cheap and asks for GCash upfront. Its profile picture is the Ghostline ghost from the café stranger's laptop.
Safe	Bought from the official store through checkout.
Recovered	Opened the too-cheap seller's chat but backed out.
Compromised	Paid gadget.deals.ph upfront by GCash; the item never came.
Lesson	Check reviews and account age, be wary of prices far below market, and use the platform's checkout or cash on delivery.
Source	FTC, "Buying From an Online Marketplace"; PNP-ACG: online selling scams were the most common cybercrime in 2024–2025

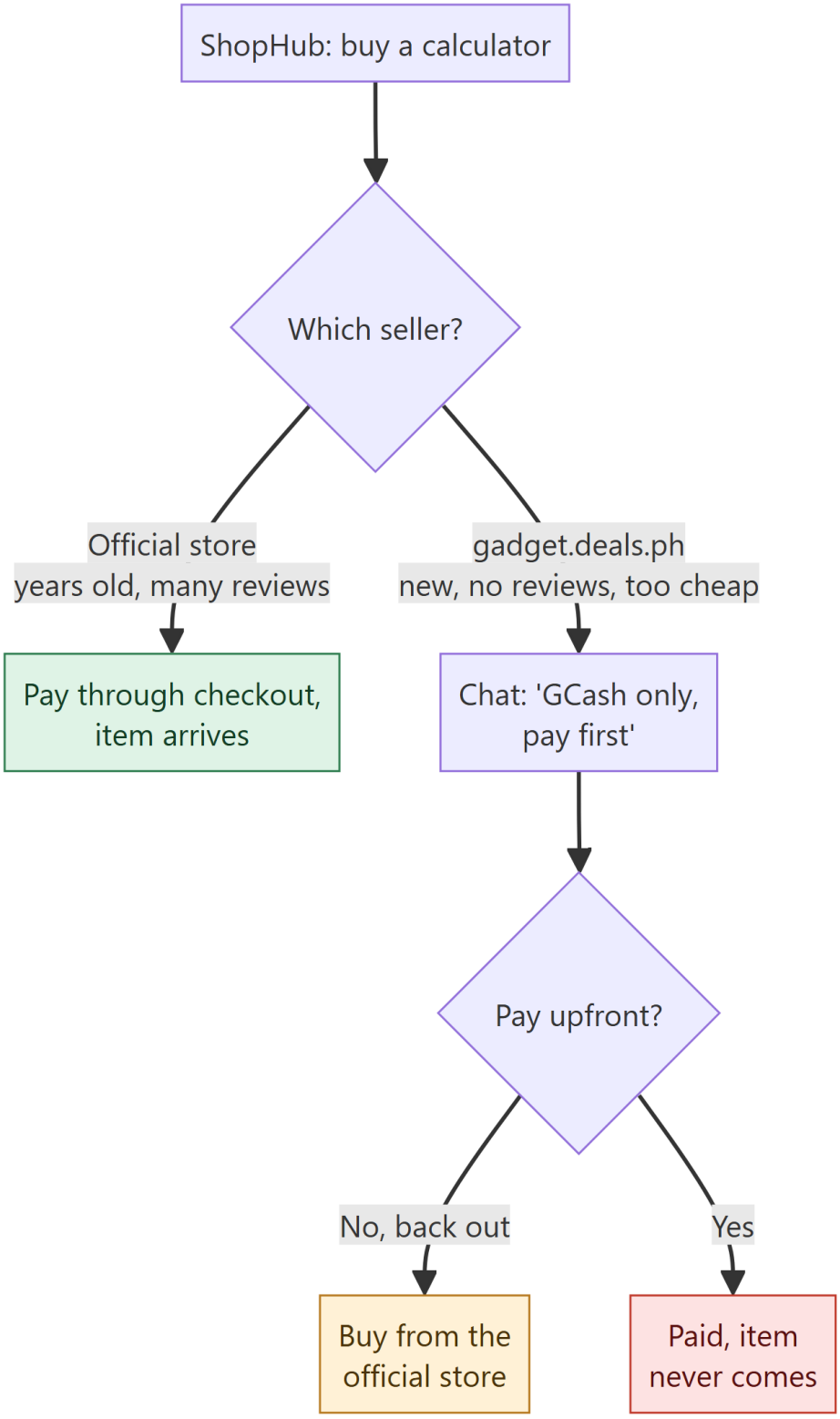


Figure 4.20 Flowchart of Fake online seller